

Task 2: - Common Examples of OS Security: -

As we mentioned in previous task, security is concerned with Attacks against: -

1. Confidentiality
2. Integrity
3. Availability

In this lesson we focus on 3 weaknesses targeted by malicious users: -

1. Authentication and weak passwords
2. Weak file permissions
3. Malicious programs

Authentication and weak passwords: -

Authentication is out of verifying your identity, be it local or remote system. Authentication

can be achieved via 3 main ways: -

- * Something you know, such as password or PIN code.
- * Something you have, such as phone number via which you can receive SMS message.
- * Something you are, such as a fingerprint.

Since passwords are most common form of authentication, they are also the most attacked.

Many users tend to use easy to guess passwords or same password on many websites.

Moreover, some users rely on personal details such as date of birth and name of their pet.

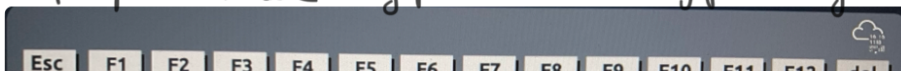
The National Cyber Security Centre (NCSC) has published a list of 100,000 most common passwords. Let's look at top 20 passwords.

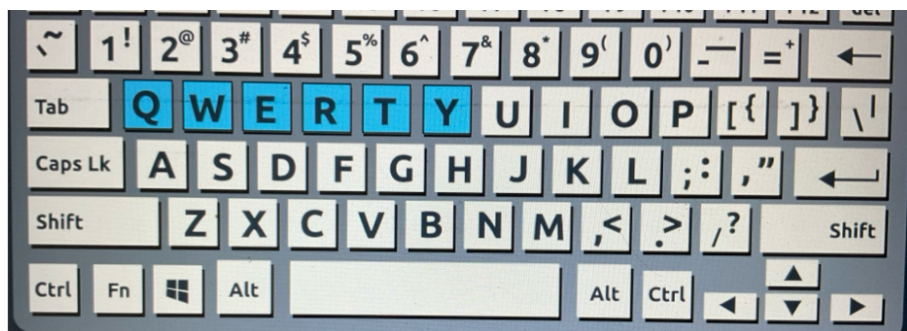
Rank	password	Rank	password
1	123456	11	1234567890
2	123456789	12	123123
3	qwerty	13	000000
4	password	14	iloveyou
5	111111	15	1234
6	12345678	16	1qaz!@WSX
7	abc123	17	qwertyuiop
8	1234567	18	123
9	password1	19	monkey
10	12345	20	dragon

We can see that 123, 1234, 12345, ..., 123456789, and 1234567890 are on the list. Dictionary words such as password, iloveyou, monkey and dragon are commonly used.

Words not in dictionary include qwerty, qwertyuiop, and 1qaz!@WSX, these seemingly

complex passwords are very predictable as they follow keyboard layout.





In brief, the attackers can guess the password of any of your online accounts, such as your email or Social media account, they will be able to gain access to your private data. Therefore it's vital that you choose complex passwords and use different passwords with different accounts.

Weak file permissions :-

Proper security dictates the principle of least privilege. In work environment, you want any file Accessible only by those who need to access it to get work done.

Ex:- on a personal level you are planning for trip with family or friends, you want to share all files related to trip plan with those going on that trip; you don't want to share such file publicly. That's principle of least privilege, or in simple terms 'who can access what?'

Weak file permissions make it easy for adversary to attack Confidentiality and Integrity.

They can attack Confidentiality as weak permissions allow them to access files they should not be able to access. Moreover, they can attack Integrity as they might modify files that they should not be able to edit.

Access to Malicious programs :-

The last example we will consider is the case of malicious programs. Depending on type of malicious program, it can attack Confidentiality, Integrity and availability.

Some type of malicious programs, such as Trojan horses, give the attacker access to your system. Consequently, the attacker would be able to read your files or even modify them.

Some type of malicious programs attack availability. One such example is Ransomware. Ransomware is a malicious program that encrypts user's file.

Encryption makes files unreadable without knowing the encryption password. So other words files become gibberish without decryption (reversing the encryption). The attacker

offers user ability to restore availability, i.e. regain access to their original files: they would give them encryption password if user is willing to pay the "ransom" (money or bitcoin).

Quiz:-

which of following is strong password, in your opinion?

- * i love you
- * learnMoox
- * 1q2w3e4r5t
- * qwertyuiop

A) learnMoox